

Project 7 Deliverables

Before Coding Period

- Implement Pinia state management for Verilog terminal
- Verify no dropped messages under rapid repeated synthesis

Week 1 (May 25 – May 31)- Error Handling & Timeout Guard

- Wrap `synthesizeVerilog()` in `AbortController` with 30s timeout via `Promise.race()`
- Add VFS guard: check `output.json` existence before parsing
- Map silent Yosys crashes → human-readable syntax error messages

Week 2 (June 1 – June 7)- Worker Lifecycle & Layout Engine

- Worker Lifecycle Management: `worker.terminate()` after N runs + dynamic respawn

Week 3 (June 8 – June 14)- Progress UX

- Move `yosys2digitaljs` inside web worker to prevent UI lag at the time of large/complex json

Week 4 (June 15 – June 21)- isTauri() routing

- Implement `isTauri` for routing

Week 5 (June 22 – June 28)- Global Removal & Reactive Terminal

- Remove all `document.getElementById` and `window.verilogTerminal` globals from `VerilogEditorPanel.vue`
- Bind terminal output, status, errors reactively to `useSynthesisStore()`

Week 6 (June 29 – July 5)- Midterm Evaluation

- Finalize Verilog2CV.js routing: isTauri=true → WASM pipeline; isTauri=false → Rails HTTP POST
- Both paths converge on identical DigitalJS circuit object; zero disruption to web users

Week 7 (July 6 – July 12)- Offline Mode & Tauri Commands

- Bundle 15MB YoWASP into /public for installer; async load from local filesystem on startup

Week 8 (July 13 – July 19)- Production Build Validation

- Validate full offline synthesis, WiFi off, no server round-trips, Windows release build

Week 9 (July 20 – July 26)- Vitest Pipeline Tests

- Full pipeline tests: progress callbacks, error handling, valid synthesis output structure

Week 10 (July 27 – Aug 2)- Parity Tests

- Deep-equality normalization: sort connector arrays by stable key, strip env metadata
- Parity tests for 5 circuits using toMatchObject(): AND gate, D flip-flop, full adder, 4-bit counter, 2:1 mux
- Confirm WASM-to-server topological equivalence for all 5 reference circuits

Week 11 (Aug 3 – Aug 9)- E2E Web Platform Safety

- Cypress E2E for web target (isTauri=false): mock HTTP 200 Rails response, verify fetch() path never boots WASM worker

- Author docs/client-side-synthesis.md: Worker boundary, store API, Tauri commands, contributor instructions

Week 12 (Aug 10 – Aug 24)- Final Evaluation

- Run oxlint, rebase branches, prepare final combined PR with
- Final milestone: Address mentor review, transition for maintainer merge